Thermodynamic principles beyond physical systems-II

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This is part-II of a set of two articles that aims at developing an abstraction of the thermodynamics principles. Part-I helped harmonize the use of the concept in thermodynamics of physical systems, in statistics of artificial systems, in information theory etc. The harmonization was not just in the interpretation of entropy and it's maximization, but also in the use of Boltzmann's distribution to describe an entropy maximized state in a constrained scenario. In this part, we push the harmonization further. If statistical physics is indeed an internally consistent abstract idea, other ideas from thermodynamics such as heat flow and equilibration should also be mappable to statistical systems like a system of dice. This harmonization is what we will explore in this part.

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I. INTRODUCTORY REMARK

As we did in part-I, we will first develop an abstract notion of thermodynamic ideas of heat transfer and equilibration and then apply it to our dice problem. We will see how the second law of thermodynamics is simply a statement of maximally likely behavior in constrained systems. In Tab. I we summarize the notation/concepts from part I. We will use the same language in this part.

II. STATISTICAL UNDERSTANDING OF HEAT TRANSFER AND EQUILIBRATION

We already witnessed in part I that the entropy functional has an additive property. That is, if the system is partitioned into two independent parts R,B, then by writing the probability p_i as a joint probability $p_j^R p_k^B$, where p_j^X is the distribution of the subsystem $X \in \{R,B\}$, we have $\tilde{S}_{R+B} = \tilde{S}_R + \tilde{S}_B$, irrespective of if $\{p_j^X\}$ correspond to the maximum entropy distribution.

Now consider two systems R and B that are ergodic (have a tendency to equally sample their configurations). They are individually in equilibrium (maximum entropy state) such that R is sampled for Σ_R^0 and is characterized with parameter β_R^0, and B for Σ_B^0 with parameter β_B^0. The combined system has the sum: $\Sigma_{R+B} = \Sigma_R^0 + \Sigma_B^{0\,1}$. But for the combined system, the sampling filter, by construction, is still only sensitive to the value Σ_{R+B} and not to the identity associated with R or B. Thus, for the ensemble of the R+B system, we are interested in the most likely sampling that will yield a $\Sigma_{R+B} = \Sigma_R^0 + \Sigma_B^0$.

object/concept	symbol/meaning
$\tilde{S}(\{\tilde{p}_i\})$	Entropy functional with undetermined probabilities $\tilde{p}_i$
S	Maximized entropy, consistent with imposed constraints
V_n	Value of individual components
V	Value for the whole system
β	Lagrange multiplier establishing the maximized entropy states
$S = \beta V + \ln Z = \ln \bar{Z}$	$Z = \sum_n e^{-\beta V_n}$, $\bar{Z} = \sum_n e^{-\beta(V_n - V)}$
Equilibrium state	Maximum entropy state
Ergodic nature	Tendency to 'sample' all states equally when no constraints are imposed
Sampling	A procedure that only retains configurations consistent with constraints

TABLE I. Table summarizing the notations, relations, and concepts in the entropy principle as introduced in part-I[?] .

This most-likely-sampling in the R+B system will, quite generally, have a $\{p_j^R\}$ and $\{p_k^B\}$ that are different from those prior to mixing them into one ensemble. The reason for this is the following. An ergodic system in equilibrium (maximized entropy) has *one* Lagrange multiplier β for the whole system. When we mix the two subsystems R and B, each at their individual β_R^0, β_B^0, the ensemble is necessarily not in equilibrium. Because both R and B are ergodic, their independence ensures the ensemble to also be ergodic, thus maintaining its tendency to maximize entropy. Subjecting it to a constraint, here $\Sigma_{R+B} = \Sigma_R^0 + \Sigma_B^0$ will, by construction, lead to a new equilibrium (maximum entropy state) with a single multiplier β such that upon mixing $\beta_R^0 \to \beta$ and $\beta_B^0 \to \beta$. Since a given Σ_R fixes a β_R and vice-versa, we should also have $\Sigma_R^0 \to \Sigma_R$ and $\Sigma_B^0 \to \Sigma_B$, where Σ_X is the new value of the subsystem X. These new values are such that $\Sigma_R^0 + \Sigma_B^0 = \Sigma_R + \Sigma_B$. This can also be expressed as

$$\Sigma_R = \Sigma_R^0 + \Sigma_Q, \;\; \Sigma_B = \Sigma_B^0 - \Sigma_Q, \tag{1}$$

where Σ_Q is the transfer of the average value to/from the R system from/to the B system, and can be positive or negative. We will refer to this as the 'heat value'. This setup (the initial conditions $\beta_R^0, \Sigma_R^0, \beta_B^0, \Sigma_B^0$) begs the following three questions

Q1 What is the final β?

Q2 What is the amount of heat value Σ_Q?

Q3 What is the relation between $\{S_R^0, S_B^0\}$, the individual initial entropies of the subsystems, and the final S_{R+B}?

To address these questions, let us start by exploring the relations at their individual equilibria. From Tab. I, we have the following expressions for the individually maximized entropies:

$$\begin{aligned} S_R^0 &= \beta_R^0 \Sigma_R^0 + \ln Z_R^0, \;\; Z_R^0 = \sum_n^R e^{-\beta_R^0 V_n^R}, \\ S_B^0 &= \beta_B^0 \Sigma_B^0 + \ln Z_B^0, \;\; Z_B^0 = \sum_m^B e^{-\beta_B^0 V_m^B}. \end{aligned} \tag{2}$$

The equilibrium of the mixed ensemble should have

$$S_{R+B} \;=\; \beta \Sigma_{R+B} + \ln Z_{R+B}. \tag{3}$$

Because $\Sigma = \sum_n p_n V_n$, this is an additive property across different subsystems. Thus, $\Sigma_{R+B} = \Sigma_R + \Sigma_B$. Because $\{p_n^R\}$ and $\{p_m^B\}$ are independent, their joint probabilities are $\{p_n^R p_m^B\}$ which leads to the Z-functions being multiplicative $Z_{R+B} = Z_R Z_B$. We already argued on general grounds that

$$S_{R+B} = S_R + S_B.$$

From the above discussion, we can identify

$$\begin{aligned} S_R &= \beta \Sigma_R + \ln Z_R, \;\; Z_R = \sum_n^R e^{-\beta V_n^A}, \\ S_B &= \beta \Sigma_B + \ln Z_B, \;\; Z_B = \sum_m^B e^{-\beta V_m^B}. \end{aligned} \tag{4}$$

This takes the same form as in Eq. (2), but this time with a common multiplier β and a different Σ for each subsystem. Now consider

$$\begin{aligned} S_R - S_R^0 &= \beta(\Sigma_R^0 + \Sigma_Q) - \beta_R^0 \Sigma_A^0 + \ln Z_R - \ln Z_R^0 \\ &= (\beta - \beta_R^0)\Sigma_R^0 + \beta \Sigma_Q + \left\{ \underbrace{\partial_\beta \ln Z|_{\beta_R^0}}_{-\Sigma_R^0} (\beta - \beta_R^0) + \frac{1}{2} \underbrace{\partial_\beta^2 \ln Z|_{\beta_R^0}}_{M_R^{[2]}} (\beta - \beta_R^0)^2 + \ldots \right\} \\ &= \beta \Sigma_Q + \frac{M_R^{[2]}}{2} (\beta - \beta_R^0)^2 + \ldots, \end{aligned} \tag{5}$$

where the ... represents higher order terms in $\beta - \beta_R^0$. The underbraces are derived easily by performing explicit derivatives:

$$\begin{aligned} Z &= \sum_n e^{-\beta V_n} \\ \Rightarrow \partial_\beta \ln Z &= -\sum_n p_n V_n = -\Sigma. \qquad (6) \\ \partial_\beta^2 \ln Z &= \sum_n p_n (V_n - \Sigma)^2 \equiv M^{[2]}. \qquad (7) \end{aligned}$$

The same applies to subsystem B, but with $\Sigma_Q \to -\Sigma_Q$. Note that the final β is the same for both subsystems. This gives

$$S_B - S_B^0 = -\beta \Sigma_Q + \frac{M_B^{[2]}}{2}(\beta - \beta_B^0)^2 + ... \qquad (8)$$

Adding the two we get

$$\begin{aligned} \underbrace{S_R + S_B - S_R^0 - S_B^0}_{\equiv \Delta S_R + \Delta S_B} &= \frac{M_R^{[2]}}{2}(\beta - \beta_R^0)^2 + \frac{M_B^{[2]}}{2}(\beta - \beta_B^0)^2 + ..., \\ \Rightarrow \Delta S_R + \Delta S_B &\geq 0, \qquad (9) \end{aligned}$$

where the sign of the right hand side is inferred from Eq. (7) which identifies $M^{[2]}$ with the variance of the distribution of V_n's and hence is non-negative definite. This result directly addresses Q3 above and was to be expected: while S_X^0 were the equilibrium (maximized entropy) states on their own, the equilibrium of the mixed state should result in an entropy that is larger. Observe that we did not explicitly impose this maximization. This got imposed in using Eq. (3) which required the resulting $\{p\}$'s to take on the Boltzmann form.

From Eq. (6), we can also infer the following relation between changes in Σ due to a change in β:

$$\begin{aligned} \Delta\Sigma &= \partial_\beta \Sigma \, \Delta\beta \\ &= \partial_\beta [-\partial_\beta \ln Z] \Delta\beta \\ &= -M^{[2]} \Delta\beta. \qquad (10) \end{aligned}$$

By definition, the change in the value of the system ($\Delta\Sigma$) is precisely Σ_Q. Thus, we also have the following relations for the two systems:

$$\begin{aligned} \Sigma_Q &= -M_R^{[2]}(\beta - \beta_R^0), \\ -\Sigma_Q &= -M_B^{[2]}(\beta - \beta_B^0). \qquad (11) \end{aligned}$$

The two equations in Eq.(11) give

$$\Sigma_Q = \frac{M_R^{[2]} M_B^{[2]} (\beta_R^0 - \beta_B^0)}{M_R^{[2]} + M_B^{[2]}}, \tag{12}$$

$$\beta = \frac{M_R^{[2]} \beta_A^0 + M_B^{[2]} \beta_B^0}{M_R^{[2]} + M_B^{[2]}}. \tag{13}$$

These equations address Q1 and Q2 posed earlier. Equation (12) also makes it clear that the sign of Σ_Q, which represents the direction of flow of value, is decided by $\beta_R^0 - \beta_B^0$. Larger β-valued system receives values.

Digression

We already saw in part-I that sampling from an ergodic system retains the property of having its entropy maximized. What we have shown here is that mixing two independent ergodic subsystems also retains the ergodic property and hence also the propensity to keep its entropy maximized. It should be noted, however, that the entropy maximization only applies to the ensemble as a whole [Eq. (9)]. The internal subdivisions into R and B are not bound by entropy maximization requirement as evidenced in Eqs. (5) and (8), where the sign of ΔS depends on $\beta\Sigma_Q$ relative to the $M\Delta\beta^2/2$ term. Thus, it is possible, and in fact common, to reduce the entropy of a subsystem in order to establish a global equilibrium. This reduction in entropy (multiplicity) of states is to be expected. We have seen in part I that $S = \ln \bar{Z}$, where $\bar{Z}$ served as a proxy for the multiplicity of the sampled states. Only in the unconstrained case does $\bar{Z}$ attain its maximum value. Constraints result in filtering of micro-states thereby necessarily reducing the multiplicity of the allowed states. When two systems mix, the need to satisfy the global constraint dictates the nature of constraint on a subsystem and that in turn dictates if the entropy (multiplicity) should increase or decrease. When this change in entropy is quantified using a value prescription, one get the notion of 'heat value' that flows from one system to the other. We see clearly that *we can access the less likely states of a system, if it can be made to equilibrate with a larger system.*

We managed to conceptualize, on purely statistical grounds, the essence of the first and second laws of thermodynamics along with many of the textbook ideas that are often just introduced as definitions. In particular, we note the following:

- The first law of thermodynamics (energy conservation) serves as the value-constraint we need to impose on the statistical evolution/sampling of our system. Due to time-translational

invariance of our physical laws, this is naturally imposed in the physical dynamics of the system, i.e. we already live in the 'sampled' state.

- The second law of thermodynamics (changes in entropy always being positive) is really a just a statement that we are only observing the most likely scenario. We have seen on very general grounds, that irrespective of what controls the evolution of systems, the statement of ergodicity ensures that all interaction between the subsystems respect the fact that the total entropy always increases.

- We have seen on abstract grounds that the approach to equilibrium is characterized by a Lagrange multiplier β and a heat flow Σ_Q. The latter is quantified in terms of the quantity $M^{[2]}$ (the second moment of a distribution) which is nothing but the heat capacity (upto a factor of β^2) introduced in thermodynamics.

- The Lagrange multiplier β maps to inverse temperature. We see quite generally that to ensure the positive change in the total entropy, the heat flow into a system (say R, which would require positive Σ_Q) happens from higher temperature (lower β_B^0) system to lower temperature (higher β_R^0) system [see Eq. (12)].

If thermodynamics has a pure statistical interpretation, as we argued, we should be able to demonstrate this in a purely statistical system such as our system of dice from part I. This is what we do next.

III. HEAT TRANSFER AND EQUILIBRATION IN THE DICE SYSTEM

Let us make our R system to be a collection of $D_R = 10$ fair red dice, and the B system to be $D_B = 10$ fair blue dice. The color is only to track the identity of otherwise indistinguishable constituents. For the individual systems, we can generate a plot for S vs Σ at equilibrium. This is shown Fig. 1 (left). This curve is obtained determining the p_i's for every bar in the histogram in Fig. 1(left) of part I. The entropy curve peaks at $\Sigma = 35$. From the histogram, it can be determined that this state has $p_n = 1/6\ \forall n$, and hence $\beta_{35} = 0$. This verifies the ergodic tendency of our individual subsystems R and B, which in turn is inherited from the fairness of individual die (see part-I[?] for discussion on this). Indeed, the expectation value for the faces in this state is $\frac{D}{6}(\sum_n s_n) = 35$. Any constraint placed on the system would deviate the Σ from 35 will necessarily reduce the entropy (the multiplicity) as shown in Fig. 1(left).

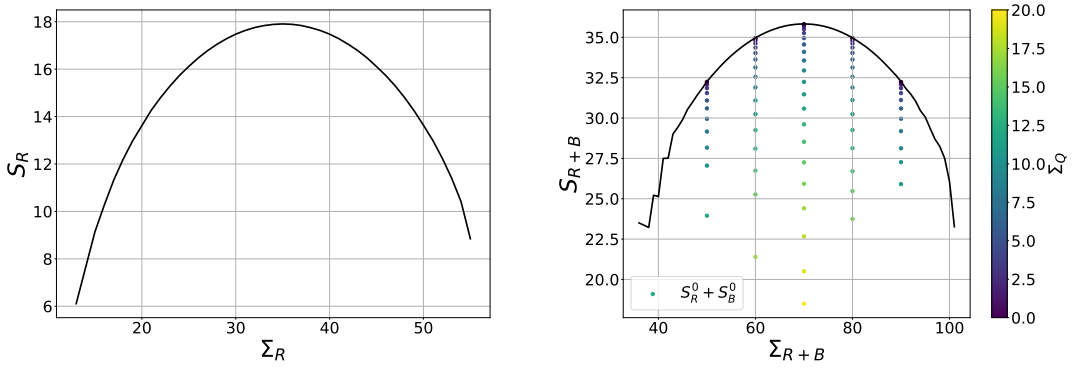


FIG. 1. [Left] The maximum entropy S at every constrained case labeled by Σ. The maximum S corresponds to $\Sigma = 35$ where all the faces are equally likely which results in $\beta = 0$. The β value is positive for $\Sigma < 35$ and negative for $\Sigma > 35$. See Fig. 2 for the β-values, but to apply it to the single system the x-axis values have to be divided by 2. [Right] The maximum entropy curve S_{R+B} (black line) for a system with 20 dice. The vertical trace are for the sum of individual entropies $S_R^0 + S_B^0$, prior to mixing, for fixed Σ's. The trace point colors correspond to the different initial conditions of mixing the R and B subsystems, characterized by the Σ_Q-value that is needed to equilibrate (i.e. reach the black curve). As the Σ_Q needed reduces, the R+B system gets closer to equilibrium. The noise at the ends is due to small frequency of occurrence closer to $\Sigma_{\min}$ and $\Sigma_{\max}$.

Now, lets mix the two subsystems. If we ignore the colors, then we are looking at a system with $D = 20$ dice and $\Sigma \in [20, 120]$ with the highest entropy state being $\Sigma = 70$. This curve is shown in black in Fig. 1(right), and is, in fact, the same curve as in 1(left) with the horizontal axis scaled by 2 to reflect addition of an identical system[2]. However, interesting scenarios can be studied if we track the colors of the dice (which essentially means we are tracking the contributions from respective subsystems). The black curve is the maximum entropy curve for a given Σ and has a well defined β-value at each Σ (see Fig. 2) which is the same for the whole system (R and B). For this value of β, the R and B subsystems will derive their corresponding Σ_R's and Σ_B's from their individual entropy curve shown in Fig. 1(left). In this particular case, because $D_R = D_B$, for the same β, $\Sigma_R = \Sigma_B = \Sigma_{R+B}/2$.

However, we could consider the scenario where $\Sigma_R^0 = \Sigma_{R+B} - \Sigma_Q$ and $\Sigma_B^0 = \Sigma_{R+B} + \Sigma_Q$. These correspond to different β-values for the R and B subsystems implying that although the subsystem R is in equilibrium with itself, and so is B, they are not in equilibrium with each other. Since

$X \in \{R,B\}$ is in equilibrium with itself, for any given Σ_X we can infer the corresponding entropy from Fig. 1(left). This allows us to get the value for $S_R^0 + S_B^0$ for different Σ_Qs. We show this in Fig. 1(right) in different color dots for select Σ_{R+B}'s. The value of $\Sigma_Q = 0$ would denote the global equilibrium (within R, within B, and across R and B), and the larger the value is, the farther is the system from equilibrium between R and B. The figure also shows that the farther the system is from equilibrium (large Σ_Q), the smaller is the entropy of the combined, but premixed, system $(S_R^0 + S_B^0)$.

In fact, the vertical traces of Fig. 1(right) can be interpreted in the following way. The bottom most point could be seen as the initial moment of mixing the subsystems. They have different β's and hence are out of equilibrium. The corresponding β's extracted from the histogram of Fig. 1(left) of Part I is plotted in Fig. 2. The approach to equilibrium is characterized by the flow of heat value from one subsystem to the other, which incrementally reduces the value of Σ_Q. Thus, although each dot on the vertical trace in Fig. 1(right) corresponds to an 'initial condition', they can be equally as well seen as intermediate steps in approaching equilibrium. We see form Fig. 1(right) and Fig. 2 that as Σ_Q reduces the entropy increases and the β-values of the two subsystems also approach the equilibrium value. Importantly, the advancement at every step, which are out-of-global-equilibrium, is also characterized by an increase in entropy. This suggests that not only the final equilibrium state is a maximized entropy state, but also that *the approach to equilibrium (in quasi-equilibrium steps) itself is governed by entropy increase*. We formalize this observation in the next subsection.

IV. THE APPROACH TO EQUILIBRIUM IN QUASI-EQUILIBRIUM STEPS

Let us break down the flow of values from Σ_R^0 to Σ_R through a sequence of steps such that after n steps we have $\Sigma_R^n = \Sigma_R^{n-1} + \delta Q$, where δQ is an infinitesimal heat transfer. For the B subsystem this would be $\Sigma_B^n = \Sigma_B^{n-1} - \delta Q$. Using the logic of Eqs. (5), (8), and (11) we can write

$$\begin{aligned} \delta Q &= -M_R^{[2]}(\beta_R^{(n)} - \beta_R^{(n-1)}); & -\delta Q &= -M_B^{[2]}(\beta_B^{(n)} - \beta_B^{(n-1)}); \\ \delta S_R^{(n)} &= \beta_R^{(n)}\delta Q + \frac{\delta Q^2}{2M_R^{[2]}}; & \delta S_B^{(n)} &= -\beta_B^{(n)}\delta Q + \frac{\delta Q^2}{2M_B^{[2]}}. \end{aligned} \tag{14}$$

$$\Rightarrow\ \delta S_R^{(n)} + \delta S_B^{(n)} \ =\ (\beta_R^{(n)} - \beta_B^{(n)})\delta Q + \frac{\delta Q^2}{2}\frac{M_R^{[2]} + M_B^{[2]}}{M_R^{[2]} M_B^{[2]}}. \tag{15}$$

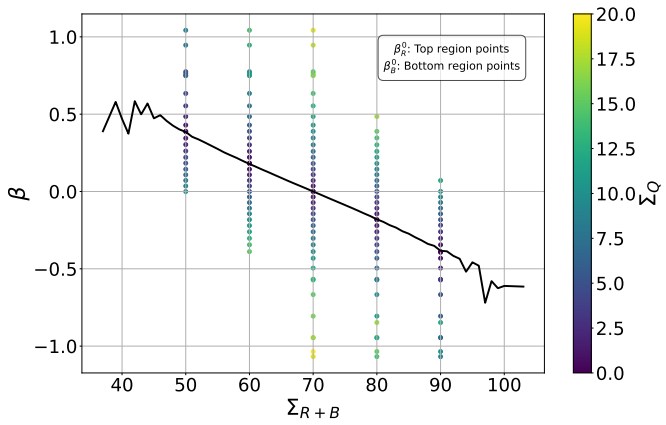


FIG. 2. The equilibrium β (black line) for the R+B system versus Σ_{R+B}. The vertical traces denote β_R^0 (above the black line) and β_B^0 (below the black line) for varying Σ_Q, which marks the departure from the equilibrium. The β-values of the two subsystems merge to the equilibrium β as $\Sigma_Q \to 0$. The noise at the ends is due to small frequency of occurrence close to $\Sigma_{\min}$ and $\Sigma_{\max}$.

For definiteness, let us say that $\beta_R^0 > \beta_B^0$ such that the ultimate transfer of value is from B to R. Since we are looking at an intermediate step, a quasi-equilibrium approach would require $\beta_R^{(n)} > \beta_B^{(n)}$ $\forall n$ (else the behavior would be erratic and push the subsystems out of equilibrium). For infinitesimal δQ, $\delta S_R^{(n)} + \delta S_B^{(n)}$ is primarily decided by $(\beta_R^{(n)} - \beta_B^{(n)})\delta Q$. Given $\beta_R^{(n)} > \beta_B^{(n)}$, the sign of the total entropy change of the R and B subsystems is entirely decided by the sign of δQ.

It is clear from Eq. (12) that when $\beta_R^{(n)} - \beta_B^{(n)} = 0$, the corresponding Σ_Q (which here is δQ) would vanish. This will stop the flow of values. At every intermediate step, because the R and B subsystems are not at equilibrium with each other, it is not actually possible to determine $\beta_X^{(n)}$ without knowing more details about the system. That is, it is not possible to determine if the movement of β_X's are monotonic or not. Here we hypothesize that in a quasi-equilibrium approach, the evolution of the β_X values should be monotonic. Coupling this with the knowledge from Sec II that the larger β-value system (here R) receives values, we have that $\forall n$, $\beta_R^{(n)} \leq \beta_R^{(n-1)}$ while $\beta_B^{(n)} \geq \beta_B^{(n-1)}$. This fixes the sign of δQ in Eq. (14) to be non-negative, and the sign of $\delta S_R^{(n)} + \delta S_B^{(n)}$ in Eq. (15) to also be non-negative. With the quasi-equilibrium hypothesis in place, we see that entropy maximization applies also to systems out-of-equilibrium, provided two self-equilibrated systems are, in fact, heading towards equilibrium by transferring values.

V. CONCLUSION

The central aim of this two-part article was to develop an abstraction of thermodynamic principles in a purely mathematical sense so as to permit its application to other statistical systems. We chose to demonstrate it with a system of fair dice. In the first part we introduced the notion of value-based states, which are derived from an underlying detailed configuration-state that is often not tractable. We introduced entropy as the log-counter of the multiplicity of value-based states. We also saw that there exists a maximized entropy state (which we labeled as equilibrium) corresponding to any specified value V and that such a state is characterized by a Lagrange multiplier β. The maximum entropy state is interesting as it is the most likely value-based observable state of the system due to its large multiplicity.

In the second part we demonstrated that when we mix two equilibrated systems, the pursuit of a new equilibrium results in a flow of values between subsystems, which we called heat. This flow can be tracked by following the evolution of the β-values for the subsystems. The new equilibrium is shown to necessarily have higher entropy than the sum of the individual entropies. That is, the mixing is the more likely outcome compared to the subsystems remaining as they were. This is an abstraction of the second law of thermodynamics. Further, consider two self equilibrated systems slowly equilibrating across each other such that they always remain equilibrated within themselves. Such an approach to equilibrium, which we call a quasi-equilibrium approach, contains out-of-equilibrium intermediate states. Despite this, a quasi-equilibrium evolution is still governed by the requirement to increase entropy. In summary, considering the two parts of the article together, we showed using a single statistical system the notion of ergodicity, entropy, equilibrium, temperature, heat flow, and the second law of thermodynamics.

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